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A New Combined Method to Enhance Adhesive Bonding of Al_2O_3 Ceramic/Aluminum by Chemical Pretreatment and Ultrasonic

Zhengwu Zhou^{1,2}, Chao Chen^{*,1,2,3}

Abstract

To effectively improve the interfacial adhesion between aluminum alloy 2024 and aluminum oxide ceramic substrates, a combined surface pretreatment and ultrasonic-assisted method was applied. Firstly, the surface of the aluminum alloy 2024 plate was roughened through hydrochloric acid etching under different concentrations and durations. The trend between joint strength and surface roughness of aluminum alloy 2024 remained largely consistent. The maximum bonding strength 27.82 MPa was achieved at a hydrochloric acid concentration of 15% and an etching time of 100 seconds. Subsequently, ultrasonic vibration was introduced during the adhesive bonding process. By integrating chemical etching with ultrasonic-assisted adhesive pretreatment, the strength of the adhesive-bonded joint was remarkably improved by 73.47%, from 18.13 MPa to 31.45 MPa. Without ultrasonic assistance, there were many flaws and defects along the adhesive-to-plates interface. However, ultrasonic assistance resulted in a void-free and uniformly distributed adhesive layer.

Keywords: Ultrasonic vibration; Hydrochloric acid etching; Adhesive bonding; Strengthening

1 Introduction

Aluminum alloy 2024 (AA2024) is an Al-Cu based alloy consisting of 92.4% Al, 4.5% Cu, and other controlled impurities. Its high strength, excellent fatigue resistance, and moderate corrosion resistance make it a preferred material for aerospace applications [1]. Aluminum oxide (Al_2O_3), with its versatile characteristics, has paved the way for a wide range of industries and applications [2]. The abundance, cost-effectiveness, and environmental compatibility of Al_2O_3 ceramic solidify its role as an indispensable structural and functional material [3, 4]. The extensive application of Al_2O_3 ceramic has led to higher performance demands on Al_2O_3 ceramic-metal composite structures [5].

Bolting or riveting cannot be used to join ceramics without compromising their performance, as such methods cause punctures in the materials [6]. Additionally, the non-reactive wetting behavior between molten metal and ceramic, due to the different chemical properties of metal (metallic bond) and ceramic, complicates bonding through soldering or brazing [7]. As a result, adhesive bonding is widely regarded as the most suitable method for joining ceramics and metal materials. While adhesive bonding of ceramics and metals has been extensively investigated [8], limited research has specifically addressed the joining of AA2024 and Al_2O_3 ceramic, an important material combination for lightweight structural applications. Our analysis of traditional adhesive joints between AA2024 and Al_2O_3 ceramic revealed inadequate adhesive to the aluminum surface. The majority of failures manifested as aluminum surface peeling, while others involved ceramic surface peeling. These findings indicate significantly weaker bonding of the adhesive to aluminum compared to ceramic. To improve joint strength and reliability, it is critical to develop techniques that enhance adhesion to the AA2024 surface.

Metal surfaces often require modification to enable robust adhesion owing to their inherent inertness and non-reactive oxide layers. A variety of chemical and physical pretreatments have been developed to overcome this challenge and prepare metallic substrates for bonding [9]. These approaches have been shown to alter surface topography [10], introduce reactive moieties [11], and ultimately transform surface chemistry to improve wettability and adhesion performance [12].

Acid etching has emerged as a versatile and precise approach for activating surfaces through controlled material removal. By corroding composites and metals in an acid solution, etching induces surface texturing that enhances adhesion [9]. This

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process generates microscopic pores and cavities across the surface, increasing the total contact area and promoting mechanical interlocking with adhesives [13]. Various acid treatments have been demonstrated to dramatically transform aluminum surfaces [14], which in turn improves the strength and durability of adhesive bonds formed with them [9]. Shokrian et al. [15] explored the effects of different aluminum surface treatments on the shear strength of adhesive bonds to identify an optimal activation approach. Their study encompassed nitric acid etching, sulfuric acid/sodium dichromate etching, hydrochloric acid etching, hydrochloric-nitric acid etching, and phosphoric acid anodization. Among these methods, hydrochloric acid etching was found to be the most effective in enhancing joint shear strength. Hydrochloric acid etching effectively roughens aluminum at both the micrometer and nanometer scales, creating a substantial interlocking area that facilitates strong adhesive bonding. To activate hydroxyapatite for adhesion, Biggemann et al. [16] developed acid etching techniques that dramatically reshape surface morphology. Immersing hydroxyapatite in hydrochloric and tartaric acid solutions etched the surface, imparting microscopic texture and enhancing roughness. This process of 'micro-texturing' not only increased the total surface area but also elevated the surface energy (especially the polar fraction), providing opportunities for both mechanical and chemical bonding with adhesives. Li et al. [17] showed that phosphoric acid anodization enables Al-Li adhesive bonding joints to maintain their strength with minimal reduction after hot-humid aging. Specifically, the anodized surface of Al-Li alloy exhibited a retention rate of 90%, unlike other treatments.

While acid etching significantly enhances the surface area by introducing porosity, this process does not inherently guarantee improved adhesion. High-viscosity adhesives often fail to fully penetrate and interact with the microscopic surface features due to their limited ability to wet porous surfaces. These adhesives are unable to anchor within microscopic cavities and channels, thereby failing to capitalize on the increased surface area and potential interlocking opportunities.

Ultrasonic assisting has recently become a focus of interest as a versatile technique for enhancing adhesive wetting and penetration [18]. When an interface is exposed to high-frequency vibrations, ultrasonic vibration efficiently decreases surface tension and disrupts boundary layers in adhesives, thus enhancing their flow across substrate features [19]. According to Wang et al. [20], ultrasonic treatment densifies the interface between adhesives and substrates, enabling far greater penetration of surface structures. Under high-frequency vibration, adhesives are capable of infiltrating microscopic pores, cavities, and imperfections that remain inaccessible under static conditions. This enhanced interlocking mechanism strengthens the joint and amplifies effective adhesion. To directly observe the effects of ultrasonic on adhesive-substrate interactions, Wang et al. [21] conducted tensile failure tests and flow visualization experiments. Ultrasonically-treated joints exhibited cohesive failure within the adhesive layer, in contrast to untreated controls, where failure occurred at the adhesive-substrate interface. The vibration-induced permeation allowed the adhesive to anchor more deeply into the substrate microstructures. Transparent plate bonding experiments further visualized the enhanced flow and dispersion of adhesives under ultrasonic stimulation. Holtmannspotter et al. [22, 23] proposed harnessing acoustic cavitation effects in adhesives to improve interface formation and enable quality assurance during bonding processes. By agitating the adhesive, cavitation promotes intimate contact and robust adhesion between substrates.

In this study, the surfaces of AA2024 plates were pretreated using hydrochloric acid etching. The impact of hydrochloric acid etching on the bonding strength of AA2024/Al₂O₃ ceramic joints was analyzed. Ultrasonic vibration was then introduced during the bonding process, and its effect on the bonding performance was evaluated. The surfaces of the AA2024 plates, the interface between AA2024 and adhesive, as well as the interface between Al₂O₃ ceramics and the adhesive were analyzed in terms of microstructure. This analysis aimed to elucidate the fundamental mechanisms underlying the observed enhancement in bonding strength.

The remainder of this paper is organized as follows. Section 2 describes the experimental materials, processing methods, and characterizations. In Section 3, the experimental results are presented and analyzed. Finally, conclusions are summarized in Section 4.

2 Materials and Methods

2.1 Materials

The ceramics used in the experiments were wear-resistant Al₂O₃ ceramics (manufactured by Shenzhen Beilong Electronic Materials Factory, China), with dimensions of 10 × 10 × 4 mm³. The aluminum alloy base material employed was AA2024 (provided by Kaili Metal Materials Ltd, China), an Al-Cu-Mg alloy that had undergone heat treatment (T3).

Laser cutting was used to process AA2024 sheets into dimensions of $80 \times 25 \times 3 \text{ mm}^3$. Figure 1 illustrates the microscopic morphology of the untreated AA2024 plate. The image shows a predominantly uniform surface, though prominent scratches are evident, which can be attributable to the initial polishing procedure.

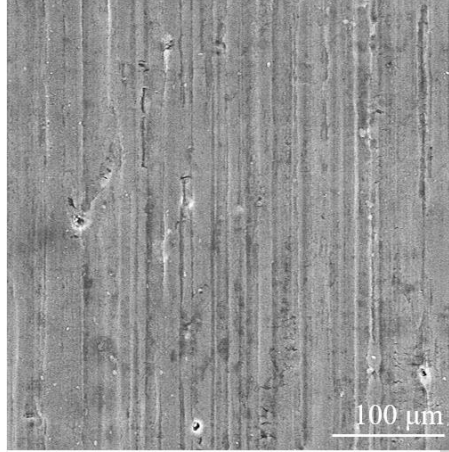


Figure 1 Microscopic morphology image of the untreated AA2024 plate

A two-part epoxy adhesive (3M DP460) was used in this study. It comprises component A and component B, which are mixed in a 2:1 ratio. The adhesive was thoroughly mixed using a mixing nozzle attached to an adhesive gun, followed by mechanical stirring to ensure homogeneity. At room temperature, the adhesive has a viscosity of $4.5 \text{ Pa}\cdot\text{s}$. Once mixed, the adhesive remains workable for 60 minutes and fully cures within 4 hours under ambient conditions. According to ASTM D1002-72, the adhesive demonstrates a lap shear strength of 31 MPa [24].

2.2 Methods

The size of the bond line is $10 \times 10 \times 0.2 \text{ mm}$, as shown in Figure 2. The thickness of the adhesive layer was controlled using a set of clamps. Together with the fixture, the joint was secured for 24 hours to cure the adhesive completely.

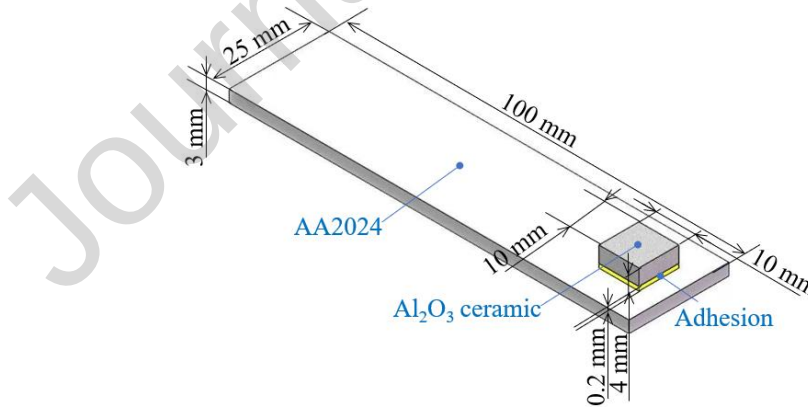
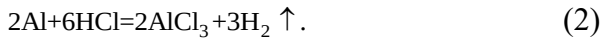
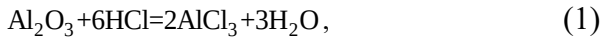


Figure 2 Adhesive-bonded joint of the AA2024/ Al_2O_3 ceramic

In this study, the etching process involved immersing the AA2024 surface in a diluted hydrochloric acid solution for a specified duration to effectively remove the oxide layer and create a clean, roughened surface that was more conducive to adhesive bonding. After etching, the AA2024 substrate and Al_2O_3 ceramic were bonded with the assistance of ultrasonic technology to enhance the adhesive strength.

2.2.1 The Process of Hydrochloric Acid Etching

The etching process employed solutions of 5%, 10%, and 15% dilute hydrochloric acid. During the hydrochloric acid treatment of an AA2024 aluminum plate, a reaction occurred between the surface of the aluminum plate and the hydrochloric acid. A large number of bubbles were observed on the surface of aluminum sheet, as shown in Figure 3. These bubbles were identified as hydrogen gas generated by the reaction. The corresponding reactions are depicted as follows:



Upon reaching the predetermined reaction duration, the AA2024 plate was immediately removed from the hydrochloric acid solution using tweezers and rinsed with a spray of deionized water to eliminate residual acid. Lastly, the AA2024 plate was submerged in deionized water and subjected to ultrasonic cleaning for 5 min. The substantial presence of black turbidity in the ultrasonic cleaner proved the effectiveness of ultrasound in removing reaction byproducts from the surface of the AA2024 sheet.

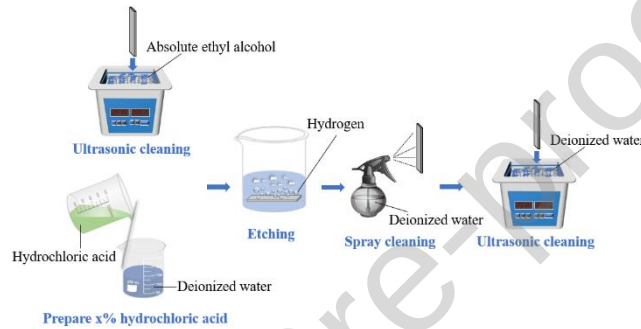


Figure 3 Hydrochloric acid-etched AA2024

2.2.2 Ultrasonic Vibration-Assisted Adhesive Bonding

After adhesive bonding between the two substrates, an ultrasonic vibration horn was placed on the surface of the AA2024 plate to transmit high-frequency mechanical waves through the adhesive layer (as illustrated in Figure 4). The workbench of ultrasonic adhesive bonding consists of an air pump, an ultrasonic generator, an amplitude transformer, an ultrasonic transducer (longitudinal), and an ultrasonic vibration horn. The ultrasonic machine outputs a frequency of 20 kHz, with the ultrasonic amplitude adjustable within the range of 20–30 μm . Once the processing parameters were determined, the AA2024 plate was subjected to ultrasonic vibration by activating the ultrasonic generator and pressing the ultrasonic horn against it. Ultrasonic vibration was applied in cycles, each consisting of a 4-second vibration followed by a 2-second pause. Following the ultrasonic vibration-assisted process, the joint was completely cured by leaving it undisturbed for 24 h.

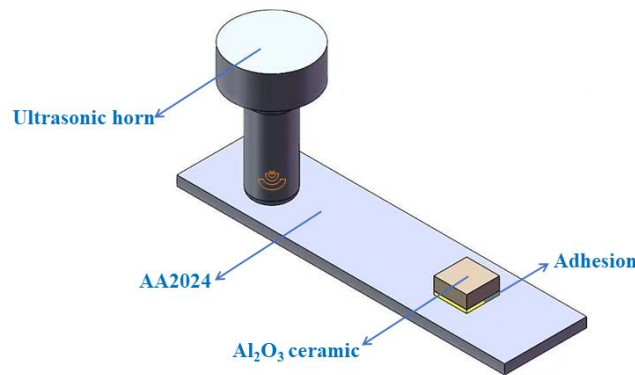


Figure 4 Ultrasonic vibration-assisted adhesive bonding of AA2024 and Al_2O_3 ceramic

2.2.3 Experimental Plan

To investigate the impact of hydrochloric acid etching pretreatment and ultrasonic application on bonding performance, three experimental groups were designed as detailed in Table 1. The surfaces of both the ceramic and the AA2024 sheet were roughened using 60 # sandpaper at first. Group A served as a control experiment, where the AA2024 surface was cleaned solely with absolute ethyl alcohol in an ultrasonic cleaner. In Group B, hydrochloric acid etching treatment was added to the cleaning process performed in Group A. For Groups A and B, adhesive bonding was conducted without ultrasonic vibration. However, for Group C, while the surface treatment remained the same as in Group B, ultrasonic vibration was employed during the bonding process.

Table 1 Experimental plan

Groups	Samples	Surface treatment	Bonding process
A	1, 2, 3, 4	Cleaning	Traditional bonding
B	1, 2, 3, 4	Cleaning and hydrochloric acid etching	Traditional bonding
C	1, 2, 3, 4	Cleaning and hydrochloric acid etching	Ultrasonic-vibrating bonding

2.3 Characterization

2.3.1 Bonding Strength Test

Shear testing was conducted to assess the mechanical properties and measure the strength of the joint. The shear test referenced common methods for aluminum and ceramic joints [7, 25, 26]. Figure 5 depicts the diagram of the shear experiment. In this setup, the joint was secured to the fixture using an Al_2O_3 ceramic, with the width of the pressing block set at 25 mm. A shear force was then applied to the joint via the side of the AA2024. During the test, the Sust CMT-5105GJ electronic universal testing machine displaced the AA2024 downward at a constant rate of 0.5 mm/min until the joint failed completely. To calculate the shear strength, the maximum force (P) and the shear area (A_s) were utilized. The shear strength was thus determined by the formula P/A_s .

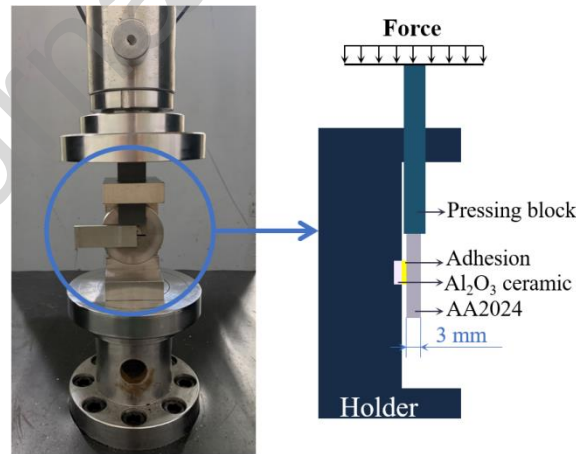


Figure 5 Shear strength test

2.3.2 Morphology of Surface and Interface

There are two methods for characterizing the surface of the AA2024:

(1) Scanning electron microscope (SEM): AA2024 samples of 10×10 mm, both before and after etching, were analyzed using SEM to investigate the impact of acid treatment on surface morphology. Ultrasonic cleaning was performed prior to imaging to remove contaminants, providing a clear visualization of texture changes induced by etching. The bonding joints

were sectioned from the position shown in Figure 6. After ultrasonic cleaning and gold coating of the bonding interface, the samples were examined using the TESCAN MIRA3 LMU field emission scanning electron microscopy (FE-SEM) system to evaluate the effect of ultrasonic vibration on the bonding interface.

(2) Atomic force microscope (AFM): For the AFM analysis, 10×10 mm samples were prepared and examined using a three-dimensional white-light profilometer (WYKO NT9100, Veeco Metrology Inc, USA). This was conducted to evaluate the surface morphology and roughness of the AA2024 plates that underwent etching treatments. Each surface of specimen was measured at three different locations, and the average values were calculated as the final results. The roughness parameters and corresponding profile images were recorded and compared for further analysis.

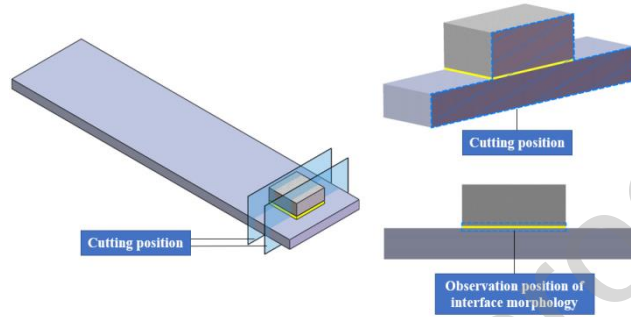


Figure 6 Cutting and observation positions for interface morphology

3 Results and Discussion

3.1 Optimization of Acid Etching and Ultrasonic Process

3.1.1 Optimization of Acid Etching

In this study, an experiment was conducted to determine the optimal combination of factors for achieving maximum surface activation. The ideal conditions were identified by systematically varying the acid concentration and exposure time across multiple levels. The specific factor levels designed for the experiments are listed in Table 2.

Table 2 The factor levels of hydrochloric acid etching experiments

Parameters	Samples									
	b-1	b-2	b-3	b-4	b-5	b-6	b-7	b-8	b-9	b-0
Hydrochloric concentration (%)	5	5	5	10	10	10	15	15	15	Untreated
Etching time (s)	50	100	150	50	100	150	50	100	150	

Based on the experimental design presented in Table 1, an experimental scheme was devised and is detailed in Table 3. Each case within the scheme was repeated four times to ensure statistical reliability. The experimental results (as shown in Table 3 and Figure 7) demonstrate a high sensitivity of bonding strength to changes in the influencing factors. Measurements were conducted with high precision (uncertainty ≤ 1.9 MPa, equivalent to 7.7% of the measured values), thereby providing robust support for the conclusions drawn. According to the data reported in Table 3, experimental condition 8 yielded the highest bonding strength at 27.82 MPa. Under this condition, the hydrochloric acid concentration was 15%, and the etching time was 100 s.

Table 3 Experimental scheme and corresponding results

Number	Hydrochloric concentration (%)	Etching time (s)	Average bonding strength (MPa)	Standard deviation
b-1	5	50	23.67 ^{+0.78} _{-0.59}	0.5752
b-2	5	100	26.78 ^{+1.77} _{-0.90}	1.2613
b-3	5	150	27.01 ^{+0.77} _{-0.88}	0.6786
b-4	10	50	24.72 ^{+1.90} _{-0.81}	1.2702
b-5	10	100	25.24 ^{+0.56} _{-0.30}	0.3942
b-6	10	150	25.70 ^{+1.31} _{-1.29}	1.3244
b-7	15	50	26.56 ^{+1.39} _{-1.41}	1.1433
b-8	15	100	27.82 ^{+1.80} _{-1.28}	1.3009
b-9	15	150	26.88 ^{+1.39} _{-1.55}	1.2056
Best level	15	100		

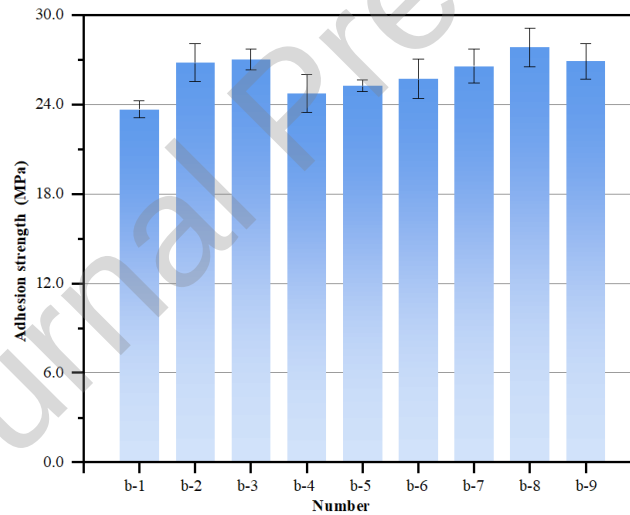


Figure 7 Bond strength under different etching parameters

3.1.2 Optimization of Ultrasonic Process

After applying the hydrochloric acid etching treatment parameters determined previously to produce maximum joint shear strength (15% acid concentration, 100-second etching time), ultrasonic vibration was introduced. The shear strength measurements were recorded for different durations of ultrasonic vibration exposure during adhesion (4 s, 8 s, 12 s, 16 s), as presented in Figure 8. According to the reported results, a steady rise in measured joint shear strength was observed with increasing ultrasonic pretreatment time from 4 s to 12 s.

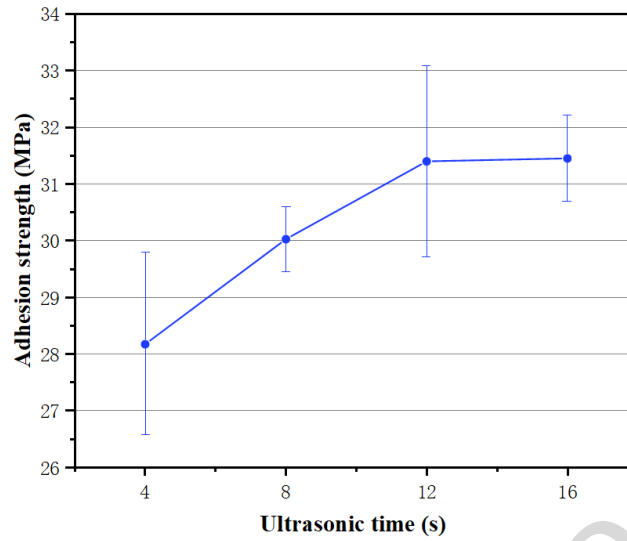


Figure 8 Adhesion strength increases with ultrasonic time

While joint shear strength improved significantly with increasing ultrasonic pretreatment time up to 12 s, it appeared to plateau and remain largely unchanged between 12 s and 16 s. This trend suggests that ultrasonic vibration may have enhanced adhesion to an optimal degree within the range of 12 s to 16 s, after which additional ultrasonic duration did not strengthen bonding significantly. There appears to be a point at which ultrasonic pretreatment maximizes its effect on shear strength by enabling sufficient surface wetting and diffusion. However, the error margin of the value at 16 s was smaller than that at 12 s. This change means that when ultrasound was applied for 12 s, although ultrasound greatly promoted bonding, the promoting effect was less consistent.

3.2 Bonding Strength and Failure Mode Analysis

Joint shear strength tests and failure mode analyses were conducted on specimens after adhesion using three different pretreatment methods. The optimal parameters obtained from the treated methods in Group B and Group C were used in this section. An examination of the fracture surfaces in joints prepared with various pretreatment methods revealed three types of failure modes:

- (1) The adhesive-bonded joint failed at the interface between the adhesive layer and the AA2024 adherend surface (Figure 9(a)). No residual adhesive was observed on the Al_2O_3 ceramic surface, indicating that the failure mode was cohesive within the adhesive layer or at the adhesive-AA2024 interface, rather than adhesive failure. This indicates a characteristic weak bond formation between the adhesive and the AA2024 surface. An examination of the adhesive layer surface revealed fine, distinct lines distributed uniformly across the surface. These lines appear to have formed as a result of the microgrooves present on the polished AA2024 plate surface before bonding [27].
- (2) In the hybrid failure area [28], a relatively thick layer of residual adhesive remained bonded to the Al_2O_3 ceramic surface (Figure 9(b)). Numerous aluminum alloy particles were extracted and cohesively held by the adhesive. These particles suggest that the adhesive has penetrated deeply into the pores of the AA2024 surface and formed a strong bond with the Al_2O_3 ceramic surface.
- (3) In Figure 9(c), the adhesive detached completely from the Al_2O_3 ceramic surface, leaving no adhesive residue behind. The absence of any remaining adhesive on the Al_2O_3 ceramic substrate suggests that the adhesive bonding had separated from the Al_2O_3 ceramic surface. This indicates an inherently weak interfacial bond between the adhesive and the Al_2O_3 ceramic.

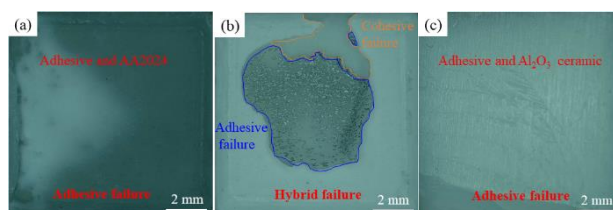


Figure 9 Failure mode after shear test

The shear strength and fracture surface characteristics of the bonds created using diverse surface modification methods are illustrated in Figure 10. Group A exhibited an average shear strength of 18.13 MPa, with failure modes classified as above (1). Hydrochloric acid etching of adherend surfaces in Group B increased the average shear strength to 27.82 MPa, representing a 53.44% improvement over Group A. The failure modes in Group B were categorized as above (2) and (3). These results indicate that hydrochloric acid etching effectively enhanced the bond between the AA2024 surface and the adhesive, eliminating adhesive failure caused by AA2024 surface issues. Group C received the same surface treatment as Group B but underwent additional ultrasonic pretreatment, resulting in an average shear strength of 31.45 MPa. This represents a 13.05% increase over Group B and a 73.47% improvement over Group A, suggesting that ultrasonic activation enhances joint strength beyond hydrochloric acid etching alone for AA2024/ Al_2O_3 ceramic bonding. All failure modes in Group C were hybrid failures. The average joint strength of 31.45 MPa achieved in Group C nearly matches the maximum reported shear strength of 3M DP460 adhesive. This indicates that optimized hydrochloric acid etching combined with ultrasonic pretreatment allows AA2024/ Al_2O_3 ceramic adhesion to approach the maximum attainable strength for the given methods and materials. Further parameter adjustments may yield only negligible improvements.

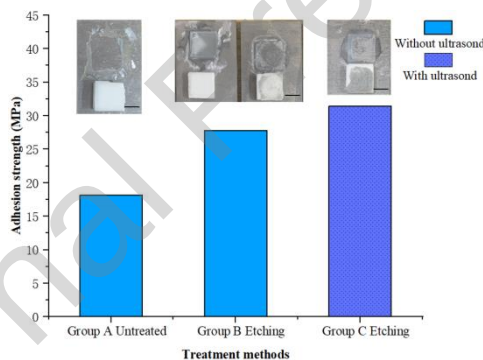


Figure 10 Adhesion strength of different treatments (Scale bar: 5 mm)

3.3 Surface Morphology Analysis

To investigate the mechanisms that enhance bonding strength, the surface topography and roughness of AA2024 plates subjected to various pretreatments were systematically examined.

SEM analysis revealed the surface structure of etched AA2024 plates (Figure 11). Compared to the untreated AA2024 plate surface (Figure 1), the hydrochloric acid-etched plate exhibited an irregular porous morphology characterized by pitting across the entire surface area. These visible surface defects were attributed to chemical reactions between the etchant and aluminum, confirming our hypothesis that etching modifies surface morphology. The changes introduced by hydrochloric acid immersion are consistent with mechanisms that promote mechanical interlocking of materials. The transition of the surface from a flat and featureless state to an irregular pattern of interconnected pores and ridges, as a result of etching, plays a critical role in enhancing the adhesion strength of coatings and bonded materials to these pre-treated AA2024 plates.

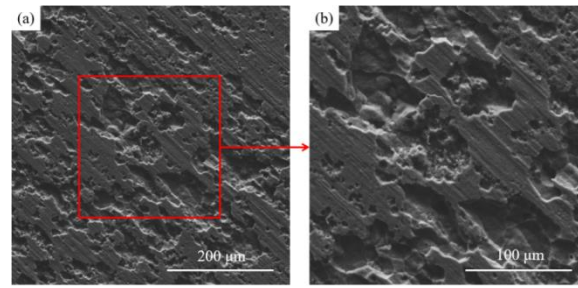


Figure 11 SEM images of AA2024 surfaces treated with different methods: (a) Cleaning, (b) Hydrochloric acid etching

Surface topographies of hydrochloric acid-etched and unprocessed AA2024 plates were characterized using AFM to quantify changes in roughness. Average values of Ra (arithmetic mean height), Rz (maximum height), Rq (root mean square height), and Rt (ten-point average height) were calculated based on multiple measurements across the sample surfaces. The measured roughness parameters are presented in Table 4. Etching significantly increased all roughness parameters relative to untreated plates. Specifically, for the untreated specimens (b-0), the values of Ra, Rz, Rq, and Rt were 0.27 μm , 4.10 μm , 0.99 μm , and 7.55 μm , respectively, corresponding to only 42.35%, 21.60%, 32.54%, and 31.18% of the maximum values observed for the etched surface (b-8, 10% concentration, 150 s). At both 5% and 10% hydrochloric acid concentrations, roughness rose progressively with etch time within the investigated range. At 15% hydrochloric acid, the roughness decreased slightly from 100 s to 150 s, which coincided with a reduction in the shear strength of bonding joints. This reversal in trend suggests that prolonged exposure to 15% hydrochloric acid may induce surface redeposition or leveling effects that counteract roughness enhancement. Figure 12 depicts the correlation between hydrochloric acid concentration, etch time, and resulting surface roughness of AA2024 plates.

Table 4 hydrochloric acid etching experiments

Parameters	Samples									
	b-1	b-2	b-3	b-4	b-5	b-6	b-7	b-8	b-9	b-0
Ra (μm)	0.30	0.34	0.43	0.37	0.38	0.39	0.45	0.65	0.56	0.27
Rz (μm)	4.11	3.59	5.1	4.25	4.83	7.21	8.95	18.98	18.78	4.10
Rq (μm)	0.95	0.68	1.25	0.68	3.05	2.71	2.72	3.03	2.56	0.99
Rt (μm)	8.93	5.38	8.73	6.84	15.50	16.61	16.28	24.21	23.08	7.55

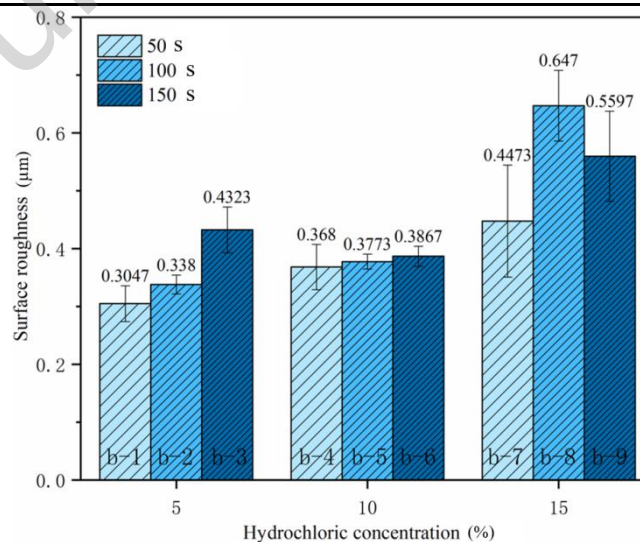


Figure 12 Effect of etching parameters on surface roughness Ra

AFM revealed distinct changes in the surface morphology of AA2024 plates in response to variable etching conditions.

Figure S1 presents three-dimensional AFM images of plates treated with different hydrochloric acid concentrations and etching times. The surface topography of an untreated control plate is shown in Figure S1 (a). Figure S1 (b)–(j) display roughness profiles of plates subjected to variable etching conditions, respectively. Relative to etched plates, the untreated surface

shows a smoother profile curve, uniform textures, and evenly distributed surface coloration, indicating lower roughness and fewer pits. Surface features of plates treated with the highest hydrochloric acid concentration (15%) studied are presented in Figure S1 (h)–(j). Compared to plates treated with lower acid concentrations (Figure S1 (b)–(g)), these plates exhibit the strongest fluctuation in profile curves, reflecting markedly increased surface roughness and a higher density of pits. A comparison of Figure S1 (b)–(d) and Figure S1 (e)–(g) indicates that as etching duration increases, so does surface roughness, as evidenced by greater fluctuations in the profile curves. Likewise, a comparison of Figure S1 (h) and (i) shows a consistent trend within this experimental group. As the etching time increased while maintaining a constant concentration of 15%, the surface fluctuations became more pronounced. The increased roughness of the profile curve indicates greater surface irregularity and height variation with prolonged etching in acid of uniform concentration. A comparison of the roughness profiles and surface images for plates etched in 15% hydrochloric acid for 100 s (Figure S1 (i)) and 150 s (Figure S1 (j)) reveals an intriguing phenomenon. Despite the longer exposure time, the surface etched for 150 s exhibited lower roughness than that etched for 100 s, as evidenced by fewer fluctuations in the profile curve. This outcome can be explained by examining the morphological changes occurring between 100 and 150 s of etching. At 100 s, the surface displayed numerous small, isolated pits that contributed to frequent instability and roughness, as demonstrated by the uneven and fluctuating profile curve. However, after 150 s, these pits coalesced, transforming the previously uneven and variably etched regions into a more uniform and flattened plane. As the surface leveled out, both instability and roughness were diminished, which is clearly demonstrated by the smoother profile curve.

3.4 Relationship between Surface Roughness and Shear Strength

To elucidate the mechanism by which etching enhances shear bond strength, the relationships between surface roughness and shear strength caused by etching treatments are presented in Table 5 and Figure 13. As shown in Figure 13, the trend in joint strength is largely consistent with the trend in surface roughness of AA2024.

Table 5 Comparison of Ra (μm) and shear strength (MPa) of AA2024 samples with different treatments

Parameters	Samples								
	b-1	b-2	b-3	b-4	b-5	b-6	b-7	b-8	b-9
Ra (μm)	0.30	0.34	0.43	0.37	0.38	0.39	0.45	0.65	0.56
Shear strength	23.67	26.78	27.01	24.72	25.24	25.70	26.56	27.82	26.88

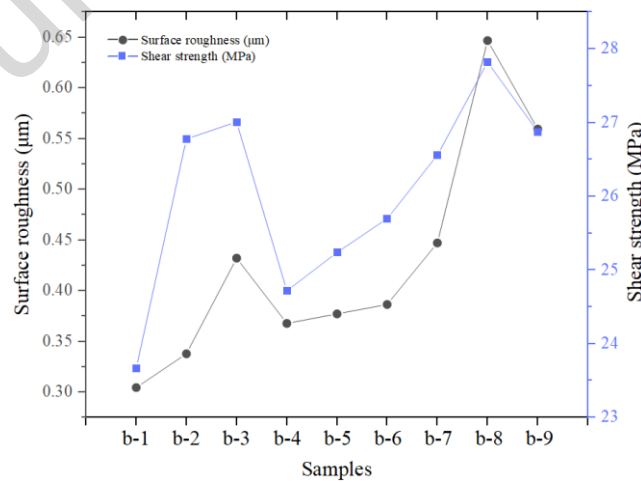


Figure 13 Relationship between surface roughness and shear strength

The dependencies can be described using the Pearson correlation coefficient method [29]. The correlation coefficient $\rho_{X,Y}$

between two random variables X and Y , with expected values μ_X and μ_Y and standard deviations σ_X and σ_Y , is defined as follows:

$$\rho_{X,Y} = \text{corr}(X,Y) = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}, \quad (3)$$

where E is the expected value operator, cov represents covariance, and corr is a widely used alternative notation for the correlation coefficient.

The closer the absolute value of the correlation coefficient is to 1, the stronger the linear relationship between the variables. The correlation coefficients between the two datasets were calculated, as shown in Table 6. Commencing with the experimental grouping protocol, samples were categorized into three groups based on hydrochloric acid concentration gradients. For the group with a 5% concentration, the correlation coefficient was 0.74479, indicating a strong linear relationship between Ra and shear strength within this group. For the groups with 10% and 15% concentrations, the correlation coefficients were 0.9992 and 0.939, respectively, meaning an extremely strong linear relationship between Ra and shear strength in these two groups. The overall correlation coefficient across all samples was 0.7575, which means a strong linear relationship between Ra and shear strength as well. It can therefore be concluded that surface roughness induced by hydrochloric acid etching significantly influences shear strength.

Table 6 Correlation coefficients between Ra and shear strength

Results	Samples								
	b -1	b-2	b-3	b-4	b-5	b-6	b-7	b-8	b-9
Ra	0.30	0.34	0.43	0.37	0.38	0.39	0.45	0.65	0.56
Shear strength	23.67	26.78	27.01	24.72	25.24	25.70	26.56	27.82	26.88
Coefficient of the same concentration	0.74479			0.9992			0.939		
Coefficient of all	0.7448								

3.5 Interface Morphology Analysis

SEM was conducted to analyze the interface between two materials that were joined together, as shown previously in Figure 6. To explore the influence of ultrasound on the joint interface, samples with and without ultrasonic treatment were analyzed. The SEM imaging results for the joint interface are presented in the current Figures 14 and 15. Specifically, Figure 14(a), (b) depict the interfacial region between the two materials for a sample that did not undergo ultrasound treatment (part of Group B samples). The interface between the adhesive and the two bonded plates was poorly consolidated, as shown in Figure 14(a) and (b). Numerous flaws and defects were observed along the adhesive-to-plates interface. These imperfections were primarily due to the inability of the adhesive to fill the surface pits and accommodate the roughness of the two plates being bonded, owing to its limited flowability. The presence of bubbles within the adhesive layer, as seen in Figure 15, would likely reduce the strength of the adhesion and the effectiveness of the bonding. As a result, the adhesive failed to establish adequate contact with the rough surfaces of the two bonded plates.

Figure 14 (c) and (d) show the interface of a sample from Group C that received ultrasonic treatment. Unlike the previous example, the adhesive here was able to spread evenly and flow over the plate surfaces, fully coating them while compensating for any surface roughness or pits. Consequently, no air pockets or areas of poor adhesive-to-plate contact were formed. The adhesive established uniform and intimate contact with the plate surfaces, leading to a solid, defect-free interface. This

bonding configuration allowed the adhesive to achieve effective anchoring and mechanical interlocking with the plate surfaces. When exposed to ultrasonic vibrations, the adhesive experienced intense and abrupt impacts against the bonded material surfaces. These impacts caused any entrapped bubbles to be expelled [21, 30].

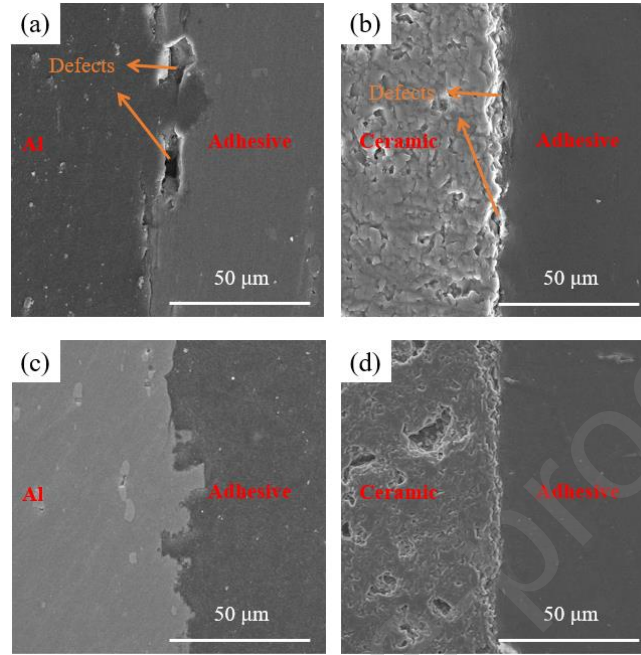


Figure 14 Morphology of the bonding interface: (a) AA2024 surface without ultrasonic vibration, (b) AA2024 surface with ultrasonic vibration, (c) Al_2O_3 ceramic surface without ultrasonic vibration, (d) Al_2O_3 ceramic surface with ultrasonic vibration

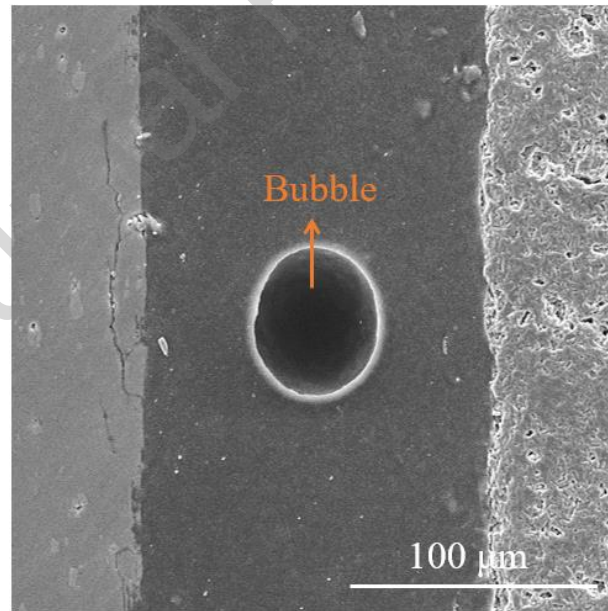


Figure 15 Bubble at the adhesive interface

4 Conclusions

- (1) The maximum adhesive bonding strength of the AA2024/ Al_2O_3 ceramic joint reached 27.82 MPa under conditions where the hydrochloric acid concentration was 15% and the etching time was 100 s.
- (2) With ultrasonic vibration pretreatment ranging from 4 s to 12 s, the bond strength increased with longer treatment times.

The strength remained stable when the ultrasonic treatment duration was extended from 12 s to 16 s. The shear strength of joints treated with both etching and ultrasonic-assisted pretreatment was 31.45 MPa, representing a 13.05% increase compared to joints treated with etching alone.

(3) Without any etching treatment, all joints exhibited adhesive failure at the AA2024 bonding surface during shear testing. For joints subjected only to hydrochloric acid etching without ultrasonic assistance, partial adhesion loss occurred at the Al_2O_3 ceramic surface, while the remaining failure area showed mixed adhesive and cohesive failure modes. When ultrasonic vibration was combined with etching treatment, all joints experienced mixed failure modes.

(4) After AA2024 plates were etched with 10% hydrochloric acid for 150 seconds, Ra, Rz, Rq, and Rt increased by 42.35%, 21.60%, 32.54%, and 31.18%, respectively. Furthermore, a strong correlation exists between the surface roughness imparted through etching and the shear strength of the AA2024/ Al_2O_3 ceramic joint.

(5) Ultrasonic vibration was found to assist adhesive in filling microscale surface defects on the etched AA2024 plates and the Al_2O_3 ceramic, enhancing the mechanical interlocking between the adhesive and the adherends. Moreover, ultrasonic treatment expelled air bubbles from the adhesive, thus improving the adhesion strength.

Authors' Contributions

Zhengwu Zhou and Chao Chen carried out the experiments and analyzed the data; Chao Chen contributed to the preparation of reagents, materials, and analysis tools; Zhengwu Zhou and Chao Chen wrote the paper. All authors read and approved the final manuscript.

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Competing Interests

The authors declare no competing financial interests.

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